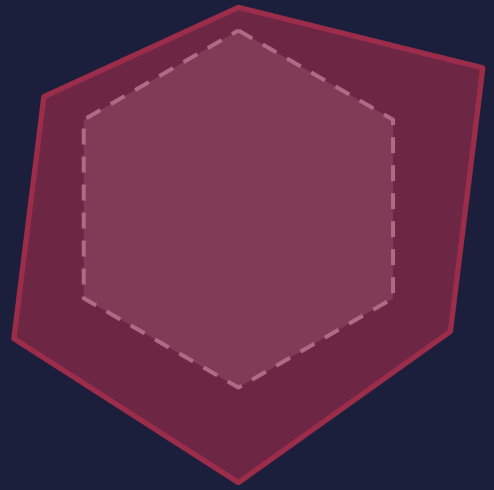
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TrueShape

Volume can change. Shape shouldn't.

A transparent monitor that flags merchants whose payments stop matching the business they registered as, and leaves shops that simply grow alone.



- - - a shop's normal shape ■ a betting front

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A betting front passes every check we run today

Genuine KYC

Real licence, real shop, correct MCC. Onboarding checks pass, and they only run once.

Every payment looks ordinary

A ₹500 UPI payment at 9 PM is normal on its own. The fraud is in the pattern, not the payment.

Honest shops change too

Festivals, a second branch or late-night delivery can triple sales. Blunt rules punish them.

104 of 150

honest shops in our test would be flagged by a plain "2× normal sales" limit.

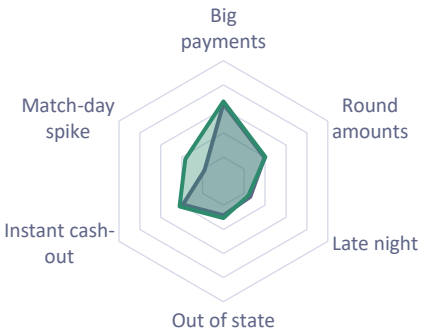
Volume rules punish growth. Careful fronts stay under the limit.

The question we answer: is this shop still running the business it registered for?

Growth changes a shop's size. Fraud changes its shape

Ovenly Bakes

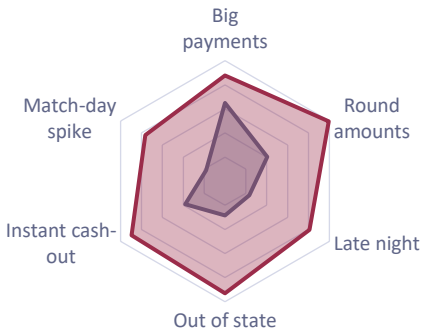
Opened a second branch



Sales 3.1× · same shape · All clear

Golden Crust Bakery

Started taking cricket bets



Shape broken · suspended in 9 days

Grey: the shop's normal weeks. Colour: its last 7 days.

Six habits make the shape

- **Payment sizes**
  Share of big and round amounts
- **Out of Shop hours**
  Payments between 11 PM and 5 AM
- **Who pays**
  Returning vs out-of-state payers
- **Where money goes**
  Cash pulled out within an hour
- **What drives sales**
  Festivals, or cricket matches?
- **Volume**
  Context only: never flags alone

No machine learning, no fraud labels: works from day one, with a plain reason for every flag.

Five checks, run every day for every merchant

1

Fingerprint

The last 7 days of payments become six habit scores.

2

Self test

Compare with the shop's own first 45 days.

3

Peer test

Subtract what similar shops in the same city did that week. Festivals cancel out.

4

Context test

Check the match calendar and the merchant's answer to "What changed?"

5

Score & act

Unexplained habits give a 0–100 score. Two must drift before any action.

Data a payment aggregator already holds

Payments (time, amount, payer) · settlement and cash-out timing · MCC and declared sales from onboarding · event calendar

→

TrueShape engine

Nightly batch in Python · plain statistics, no training · a written reason for every flag

→

Actions

Risk console for analysts · "What changed?" check-in in the merchant app · STR workflow to FIU-IND

Respond in proportion: ask before acting

Suspend & report	Account suspended, STR filed with FIU-IND
Hold funds	New payouts held while an analyst reviews
Delay payouts	Settlement moves from T+1 to T+3; shop keeps selling
Ask: what changed?	Two-tap check-in in the merchant app; money flows
Watch	Monitored quietly; the merchant sees nothing
All clear	No contact

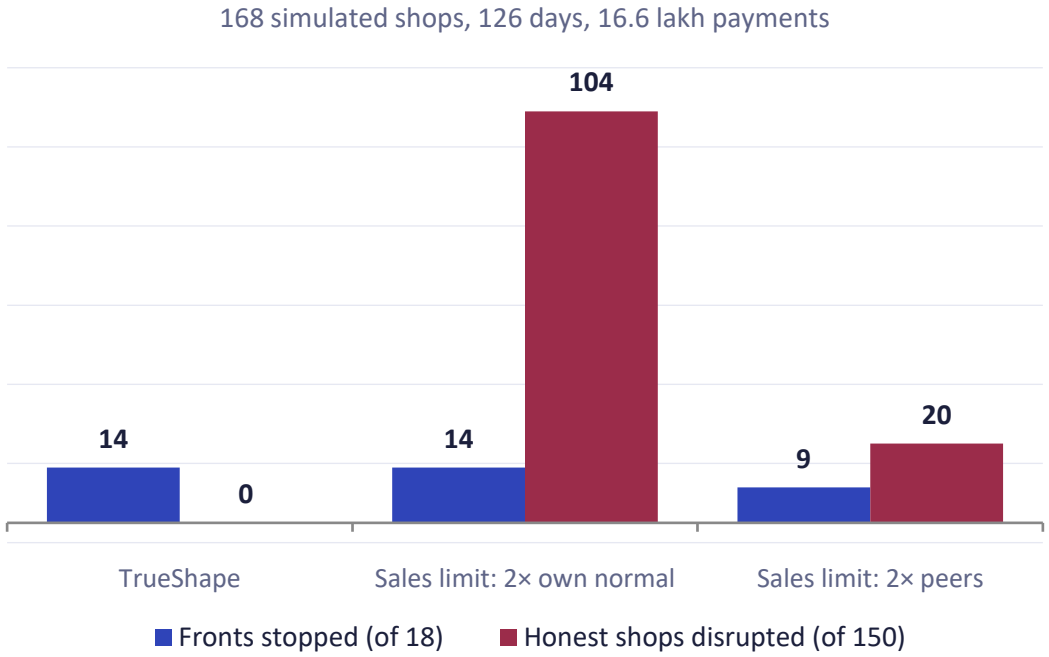
Guardrails: one drifting habit never goes past Watch. Levels step down one at a time. Only a human lifts a suspension.

Case: Golden Crust Bakery, Hyderabad

- **9 Sep** — First bets: ₹500 and ₹1,000 stakes, late at night, payers from other states
- **12 Sep** — Watch
- **13 Sep** — Check-in sent, payouts delayed
- **14 Sep** — Replies "festival rush": doesn't explain night hours, distant payers or instant cash-out. Funds held
- **18 Sep** — Suspended and reported, 9 days after the first bet

Honest shop, same rules: Midnight Cakes starts late-night delivery, answers "started late-night orders" and is cleared in 2 days. That becomes its new normal.

Fronts stopped, honest shops left alone



2–4 days

Bold fronts: from first bet to delayed payouts

16–18 days

Careful fronts that hide bets inside normal hours and odd amounts

0 of 46

Honest shops with real business changes disrupted. 6 were asked one question, then cleared

Stress test: when every front gives its most convincing excuse, the result holds at 14 stopped and 0 honest shops hit.

Built to slot into a payment aggregator's risk stack

Data

- Payment logs: time, amount, payer
- Settlement and cash-out timing
- MCC and declared sales from onboarding
- Public festival and match calendars

No new data collection

Process

- Nightly batch scores every merchant
- Analysts review only Hold cases
- Check-in via dashboard, app or WhatsApp
- STR drafted from the case file

Runs on a laptop for 168 shops in seconds

Policy and partners

- RBI PA rules: ongoing merchant monitoring
- PMLA: suspicious reports to FIU-IND
- Banks and NPCI: payer signals across providers
- Police cyber cells: front networks

Fits existing compliance duties

Rollout

1 Shadow mode
6 weeks beside current rules; tune thresholds

2 Watch + Ask
Check-ins go live; no money touched

3 Payout controls
Delay and hold for confirmed patterns

4 Scale
More categories and cities; link shops sharing payers

How we measure success, and what comes next

Success metrics

- Fronts stopped, and days from first bet to payout delay
- Honest merchants disrupted: target under 1%
- Check-in answer rate and share cleared
- Rupees held before they leave the system
- Analyst hours per confirmed front

Known limits

- Tested on simulated data we built; real thresholds come from shadow mode
- 4 careful fronts that started late stayed on Watch, not stopped
- Payer location isn't always known for in-store UPI

Next steps

- Link shops that share payers to expose betting networks sooner
- Add categories and cities with the same five checks
- Learn from analyst decisions to tune thresholds

Volume can change. Shape shouldn't.

Working prototype: TrueShape console (FastAPI + web), 168 shops, 16.6 lakh simulated payments.